

I Doubt, Therefore I Think, Therefore I Am

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"I Doubt, Therefore I Think, Therefore I Am" is an installation of three thermal printers that print an endless inner monologue onto continuous paper. The three printers stand on simple plinths on a grey office carpet - evoking associations with open-plan offices, bureaucracy, the home of the aesthetics of exploitation. Three voices of a ruminating instance interact with each other: a doubting central voice carrying the main train of thought, an escalating left voice, and a relativizing right voice. A microphone on the floor allows visitors to intervene in the stream of thought. A processed breathing sound emerging from a body-transducer lends the installation a physical presence oscillating between human and machine. An archive box, filled with crumpled receipts from previous test runs, makes the question of where produced data goes physically visible. The work addresses the structural analogy between human overthinking and the workings of large language models: both are unable to stop processing, both consume resources in the process, and both produce more doubt than answers.

1. Starting Point

The starting point of this work is a personal observation: overthinking - the circling, unending re-examination of a situation - has no off-switch. The thought returns, varies slightly, keeps circling. It consumes time, energy, and concentration without leading to a result. Whoever overthinks usually knows it leads nowhere - and continues anyway.

This experience structurally resembles the workings of large language models. An LLM like Claude Haiku, which is used in this installation, can in principle continue writing endlessly as long as it is fed with input. It has no internal reason to stop. It produces language because that is its function - not because it wants to reach a conclusion.

Here too: resources flow in, text flows out, and the actual question - is this necessary? - remains unasked.

The installation makes this double structure physically tangible. Three thermal printers, of the kind used in cafés and shops for receipts, print out a stream of thought that looks like a receipt. The receipt is a precise image for the economy in which both the overthinking subject and generative AI are embedded: both are part of a logic of exploitation in which every mental or computational stirring is protocolled, translated into paper or tokens, and accumulated. The heap of paper on the floor is the visible remainder of this process - the waste of a productivity nobody ordered.

2. Concept

2.1 Three Voices

The installation consists of three printers, each corresponding to an inner voice. The middle printer ("Middle") generates an ongoing main monologue. The left printer ("Left") reads along with every thought of the Middle and reacts by escalating - catastrophising, in drastic corporeal images. The right printer ("Right") tries to relativise the same thought, to play it down - dryly, self-ironically, without ever actually coming to rest. The three voices are tuned to one another such that Middle occasionally picks up on comments from Left or Right: the circling thereby becomes polyphonic.

The source material for the voices consists of the author's own writing: fragments from an autobiographical text, from columns, and from essays. These fragments are not reproduced verbatim by the language model but imitated - sentence structure, hesitation ("I think..."), rhetorical self-questioning. Additionally, with a low probability, machine-like doubts flow into the stream of thought: what would an AI overthink? Its own electricity consumption. Data centres in Nevada. Training data of unclear origin. Military applications. Forgetting. These moments are meant to appear unexpectedly, like a tear in the fabric of human self-doubt.

2.2 Interaction

A handheld microphone (Shure SM58) lies on the floor. Visitors can pick it up and speak into the stream of thought. The reaction unfolds in several steps: all three printers pause briefly, print the recognised sentence in inverted form (white on black) on all three paper strips – as a visual sign that a foreign thought has entered – and then begin to overthink the sentence heard. Over several cycles, attention remains on this foreign voice before the original loop resumes.

The installation's sound comes from a body-transducer (SolidDrive SD1sm) mounted on one of the plinths. Through this, the installation itself becomes the speaker: a processed breathing sound – partly the author's own recordings, partly breathing sounds generated with ElevenLabs – emanates from within the installation rather than from a separate box. The breathing stops as soon as someone speaks into the microphone, a sharp intake of breath is heard, irritation, and the breathing only resumes when the overthinking continues.

2.3 Physical Form: Plinths, Carpet, Archive

Originally, the plan was to suspend the three printers from the ceiling on transparent nylon threads – a floating that would have emphasised the distance to the everyday nature of these devices. This idea was rejected after a feasibility check: ceiling mounting in the exhibition situation would have been too technically and temporally demanding. The printers now stand on simple plinths, the paper falling from there to the floor. In retrospect, this decision proves conceptually stronger: the printers are not elevated but retain their object character as banal office devices.

The plinths stand on a turquoise office carpet. The carpet is conceptually central: it brings the language of the open-plan office into the exhibition space, the aesthetics of reception desks and waiting areas, where people have to be productive without knowing exactly for what.

Next to the installation stands an archive box – a simple box of the kind used for storing files. It is filled with crumpled receipts from previous test runs of the installation. The box poses an explicit question: where does our data go? How do we archive today? The receipts have become illegible not because language has been lost, but because sheer quantity has made them inaccessible. This is a precise description of the contemporary relationship to data: the problem is not the absence of recording, but the impossibility of meaningful review.

The title refers to Descartes' famous "Cogito ergo sum" in an extended form: "Dubito, ergo cogito; cogito, ergo sum" – I doubt, therefore I think; I think, therefore I am. Doubt is not overcome but constitutes thinking itself.

3. Technical Implementation

3.1 Hardware and Software

The system runs on a MacBook Air (M2) to which three thermal printers (Caysn, 80mm paper width, ESC/POS) are connected via a powered USB hub (Sandberg, 36W). The microphone runs via an iRig PRO interface over USB; the body-transducer is driven via the headphone output and a small amplifier.

The software is a single Python script. For each cycle, the three voices are first generated: the Middle in a first API call, Left and Right subsequently and in parallel within a shared thread pool. This allows all three printers to print almost synchronously, without the reactions of the side voices lagging behind in time. Speech recognition uses Google Speech Recognition with a fixed energy threshold, so that only direct speech into the microphone (not ambient noise or printer sounds) is recognised.

3.2 The Path from Local to Cloud Model

A first prototype of the installation used Ollama with the llama3.2 model, which runs locally on the laptop. This decision was consciously conceptually motivated: if the installation thematises the resource question

of AI, it would be consistent to work with a local model that does not use cloud infrastructure. The computational effort would be locally bounded and traceable.

In practice, however, two problems emerged. First: the interaction between the three voices was noticeably less precise than with a larger model. The escalating voice often did not react specifically to what the Middle had just thought, but produced generic catastrophising sentences. The overthinking therefore felt mechanical rather than mental. Second: the laptop became so hot with three parallel streams that a 48-hour runtime would have been questionable.

The switch to the cloud-based Claude API (Haiku 4.5) significantly improved the content quality – the voices now refer precisely to each other – but shifted the computational load into a data centre invisible from the laptop's perspective. It is precisely this contradiction that is productive: the installation now no longer shows a locally closed AI, but one that is part of a global infrastructure – and that conceals this participation. The heap of paper in the exhibition space is the only visible remainder of a process whose actual location lies elsewhere. This is conceptually sharper than the original plan.

3.3 Memory and Print Layout

The Middle's stream of thought is continuously saved to a text file. With each new API call, the last ca. 500 characters are passed along as context, so that the model does not lose the thread. If the script is stopped and restarted, it reads this file and continues where it left off.

Because the paper falls downward, the most recently printed text is always the topmost. So that a coherent thought remains readable nonetheless, each thought fragment is first generated in full, then broken into lines and sent to the printer in reverse order. This way, the thought within a paragraph reads linearly from top to bottom again. An additional rule checks whether a paragraph ends with a punctuation mark; if not, a short

additional API call is made to complete the sentence, in order to avoid breaks in the middle of a word.

4. Critical Reflection

4.1 What Succeeds, What Does Not

The work succeeds in several places. After several iterations, the three voices are differentiated enough that in their interplay they become readable as three distinct inner instances - the escalating one stays in corporeal, drastic images; the relativizing one in dry, banal formulas; the Middle in the recurring "I think...". The interaction via the microphone works reliably and creates a genuine break in the loop. The printed heap of paper works physically as an image of what it depicts: mental inability-to-stop.

Less successful so far is the complete dissolution of the foreignness of the AI material. The machine-like doubts inserted into the stream of thought ("How much electricity did I just use?") sometimes feel imposed rather than integrated - more like quotes than a voice. This is partly due to the language model itself: Claude is trained to answer helpfully and coherently. It structurally resists actually overthinking. The shorter, more disruptive moments are thereby smoothed out. It would be interesting in the future to return to the idea of a local LLM and continue experimenting with it, making this bridge productive.

4.2 Making Resources Visible

The installation is a resource machine. During a 48-hour runtime, it consumes around four to six rolls of thermal paper (240 to 480 metres, chemically coated paper that cannot be recycled as waste paper) and generates API costs on the order of 15-40 US dollars for around 15,000 to 20,000 calls to Claude Haiku - costs that translate into computing power in a data centre. The laptop runs continuously under load, the USB power supply continuously draws electricity, the printers consume

thermal energy for every printed dot. These costs are explicitly part of the work: they make the exploitation economy in which we produce both our thoughts and generative AI literally visible.

The archive box extends this question into a temporal dimension. The receipts in the box show: a lot has already been printed before this exhibition. What of it? Where did the corresponding requests of the test runs go? To a server at Anthropic, in a log file? To Google, because the speech recognition ran through them? The box makes not-knowing exhibitable. It is at the same time an archive and the negation of an archive: none of it is any longer accessible in an ordered way.

4.3 Never Being Off-Grid

Another theme: the structure of always-being-reachable. The installation permanently needs electricity, internet, USB connections. If one of these fails, it stops "thinking". In this sense, it is a direct translation of an experience of exhaustion familiar to many people from the digitised working world: those who cannot go off-grid also do not stop processing. Burnout is not a random side-effect but the logical consequence of a structure that provides no stopping.

5. Methodological Reference to the CAS

The work implements several methods and concepts taught in the CAS "AI for Creative Practices". Prompt engineering is central: the three voices are constructed through carefully elaborated system prompts that combine stylistic specifications (fragment length, forbidden formulations, concrete reference to freshly generated text) with role specifications (escalating, relativizing, doubting). Examples in the prompt are drawn from the author's own writing material, which increases the stylistic proximity to the author. Streaming APIs enable reactive interaction in near real time.

The comparison of local vs. cloud model (Ollama/llama3.2 vs. Claude Haiku 4.5) reflects one of the fundamental decisions of contemporary AI applications: the sovereignty and control of the local model versus the quality and ease of the cloud model. This decision was explicitly reflected upon and documented throughout the project. The handling of sound was additionally supported with ElevenLabs Sound Effects – the generated breathing sounds were combined with the author's own recordings to create a sonic world oscillating between human breathing and machine texture.

6. References and Sources

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